Solution Requirements

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*Users and administrators can securely log in using a username, password, or email verification.*	High
2	Authorization levels	*Residents can view flood predictions and alerts, while administrators can manage datasets, models, and notifications.*	High
3	External interfaces	*The system integrates with Weather APIs, River Water Level Sensors, and GIS/Map services to collect real-time data.*	High

4	Transactions processing	*The system collects weather and river data, processes it using the AI/ML model, predicts flood risk, and sends alerts to users.*	High
5	Reporting	*Generates flood prediction reports, historical analysis, dashboards, and alert summaries for authorities.*	Medium
6	Business rules, etc.	*The AI model classifies flood risk into **Low, Medium, or High** based on rainfall, river level, humidity, and weather conditions.*	High
7	Compliance to laws or regulations	*The system follows data privacy guidelines and government disaster management policies while handling user information.*	Medium
8	*Other*	*The system sends real-time SMS, email, or mobile app notifications during high flood risk situations.*	Medium

Step 2: Non-Functional Requirements (NFR)

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SKILL WALLET
A SmartBridge Product

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*The system should process incoming data quickly and display flood predictions without delay.*	*Prediction generated within **5 seconds***
2	Scalability	*The system should support many users and continuously process increasing weather data.*	*Supports **10,000+ users***
3	Security & Data Privacy	*User information and prediction data should be securely stored and protected.*	***AES-256 encryption** and secure authentication*
4	Reliability & Availability	*The system should remain available during emergencies with minimal downtime.*	*99.9% uptime*
5	Usability & Accessibility	*The interface should be simple, responsive, and accessible on mobile and desktop devices.*	*User-friendly interface with responsive design*
6	*Other*	*The AI model and datasets should be easy to update and maintain for better prediction accuracy.*	Easy maintenance and regular model updates